

Spatial Data Science

Graduate – Elective (MSDS/MSCS/PhDCS)



Remote Sensing
&
Spatial Analytics

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Everything is related to everything else,
but near things are more related than
distant things.

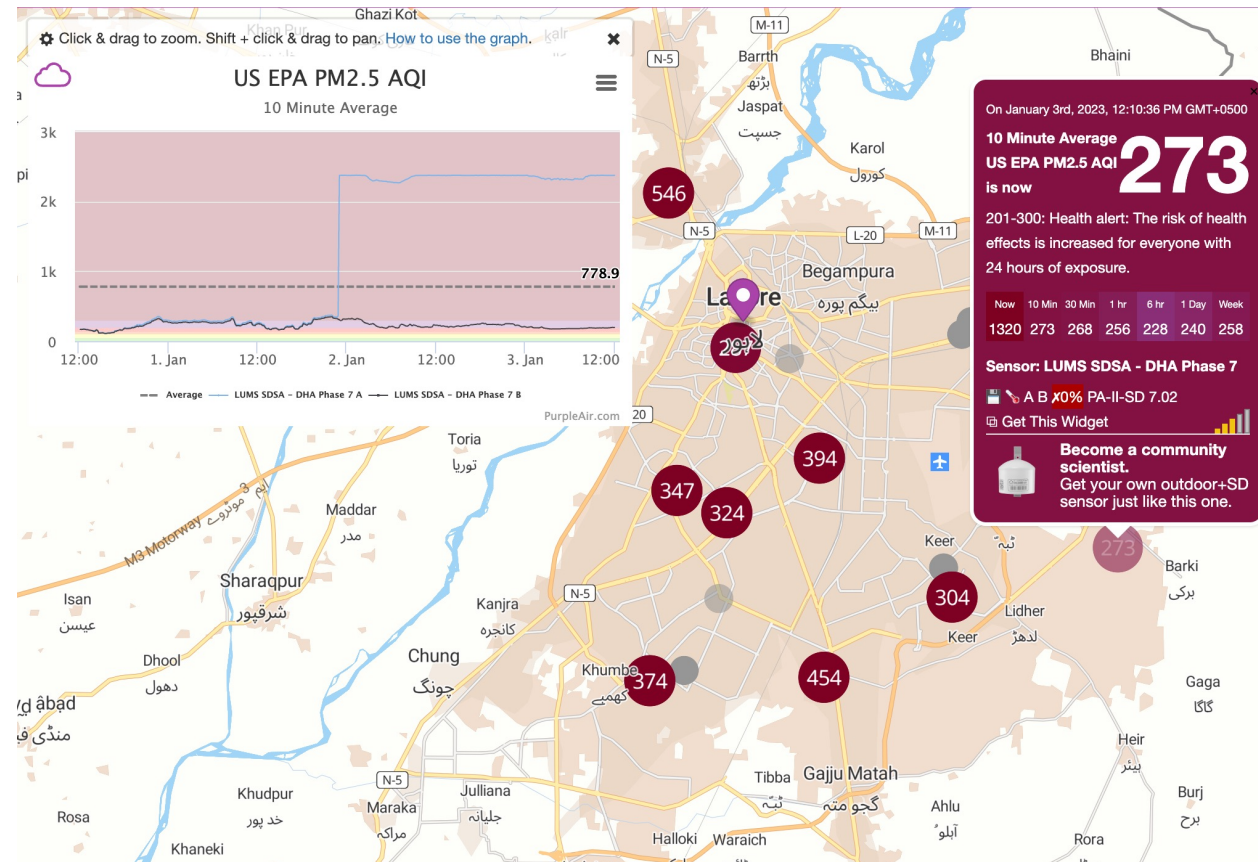
W. Tobler

Geographic thinking?

Why geographical information deserves special care and attention?

Geographic thinking for data scientists

- Spatial data is ubiquitous.
 - Everything has a location in *space-time*, and this location can be used directly to make better predictions or inferences.



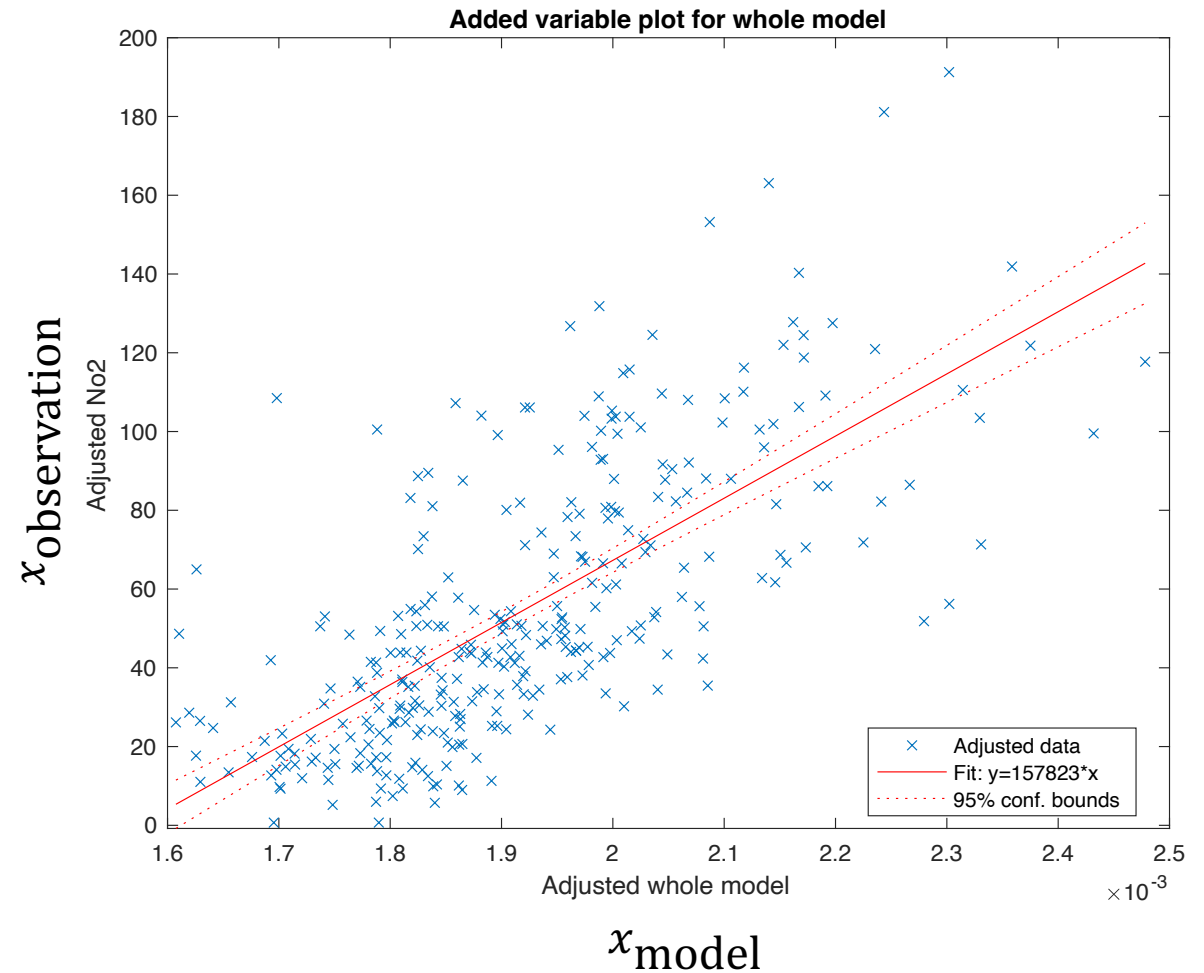
Geographic thinking for data scientists

- Spatial data is ubiquitous.
 - Everything has a location in space-time, and this location can be used directly to make better predictions or inferences.
- Location allows you to understand the relations between observations.
 - It is often the relations that are useful in data science because they let us contextualize our data, building links within our existing data and beyond to other relevant data.
- Therefore, if we learn from this contextual information appropriately, we may be able to “**build better models**”.



Models

- Models are simplified representations of reality.
- Sometime, *extra* simplified!
- Despite the fact that these representations are not exactly correct in some sense, they are useful in understanding what is important about a statistical process.



All models are wrong, but some are useful



Maps as »Models«

- A map is a representation of the underlying geographical process, but it is not the process itself.
- Despite the fact that these representations are not exactly correct in some sense, they are useful in understanding what is important about a geographical process.



All maps are wrong, but some are useful



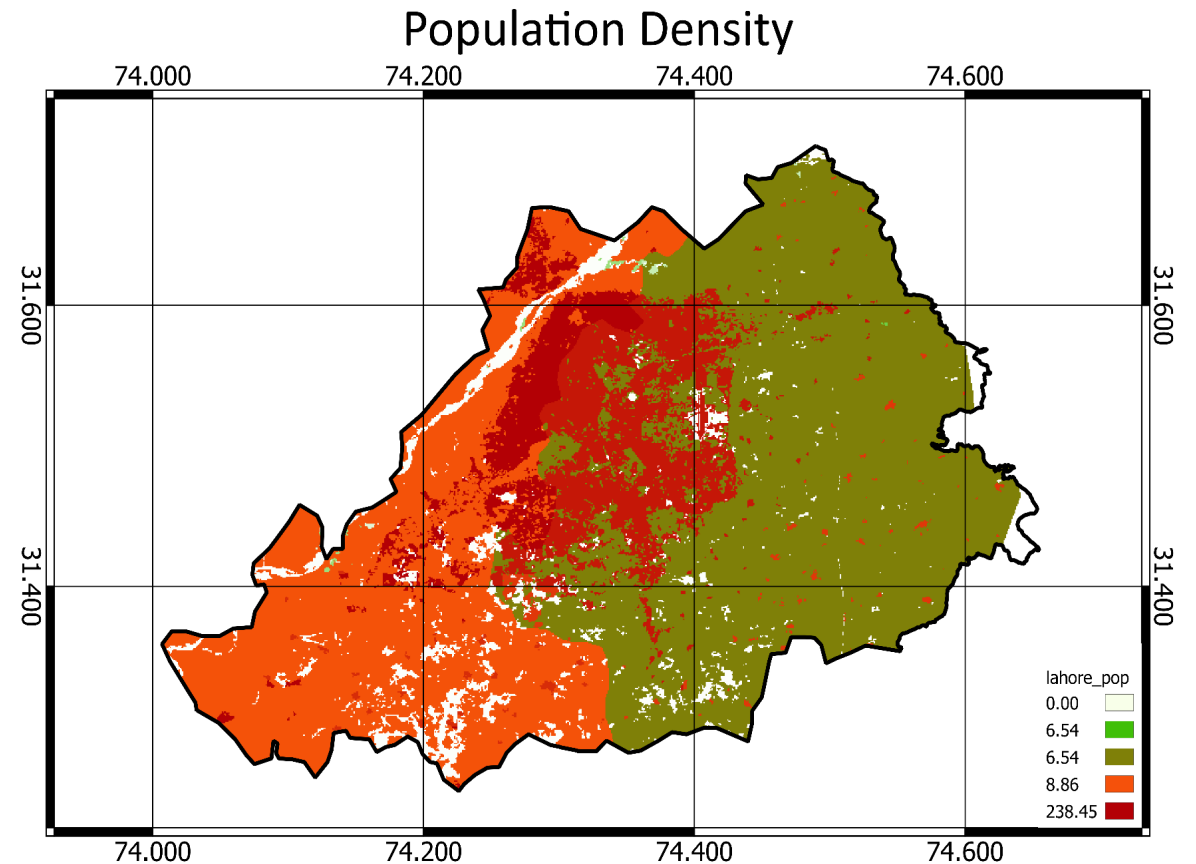
Objects, fields & networks

- *Objects* are discrete entities that occupy a specific position in space and time.
- *Fields* are continuous surfaces that could, in theory, be measured at any location in space and time.
- *Networks* reflect a set of connections between objects or between positions in a field.



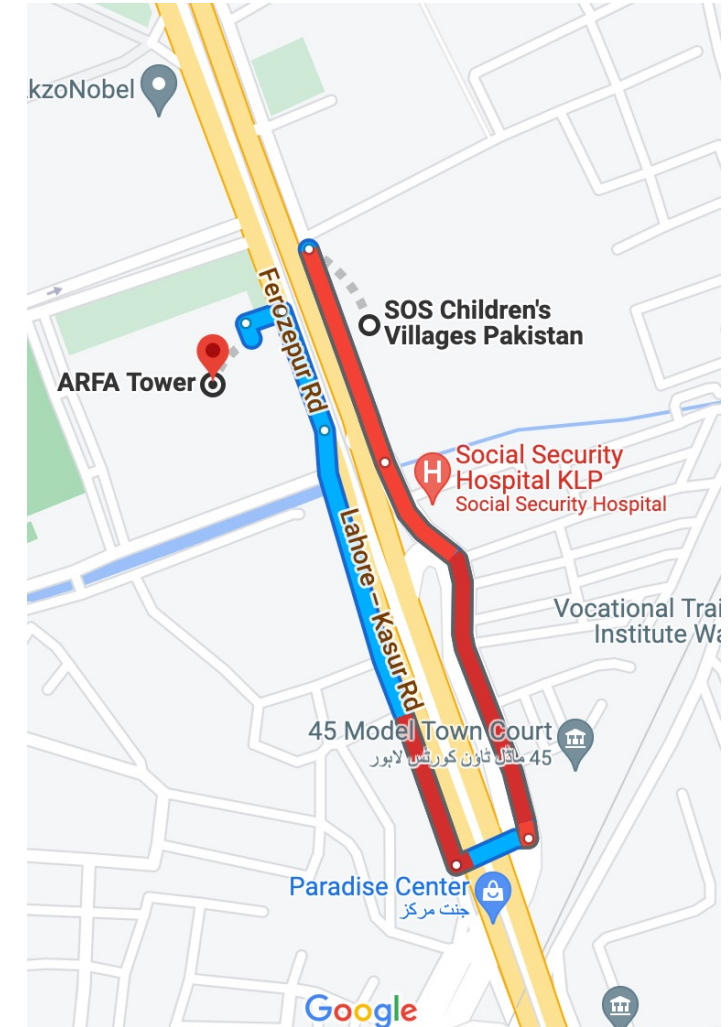
Example

- Consider population density of Lahore
- In our population density example, an “enumeration unit” is an object, as is a person.
- The field representation would conceptualize density as a smooth, continuous surface containing the total number of persons at all locations.
- The network representation would model the inter-related system of places whose densities arise from people moving around.



Relations

- The differences between these three representations are important to understand because they affect what kinds of relations are appropriate.
- For instance, the relationships among geographical processes with objects can be modelled using simple distances.
 - Near objects might then be “strongly related,” and distant objects “weakly related.”
- Alternatively, we could consider (or construct) a network that relates the objects based on their interactions.
 - e.g. for traffic management, raw distances between two points is not as strongly related as how they lie on the road network



Computational representations

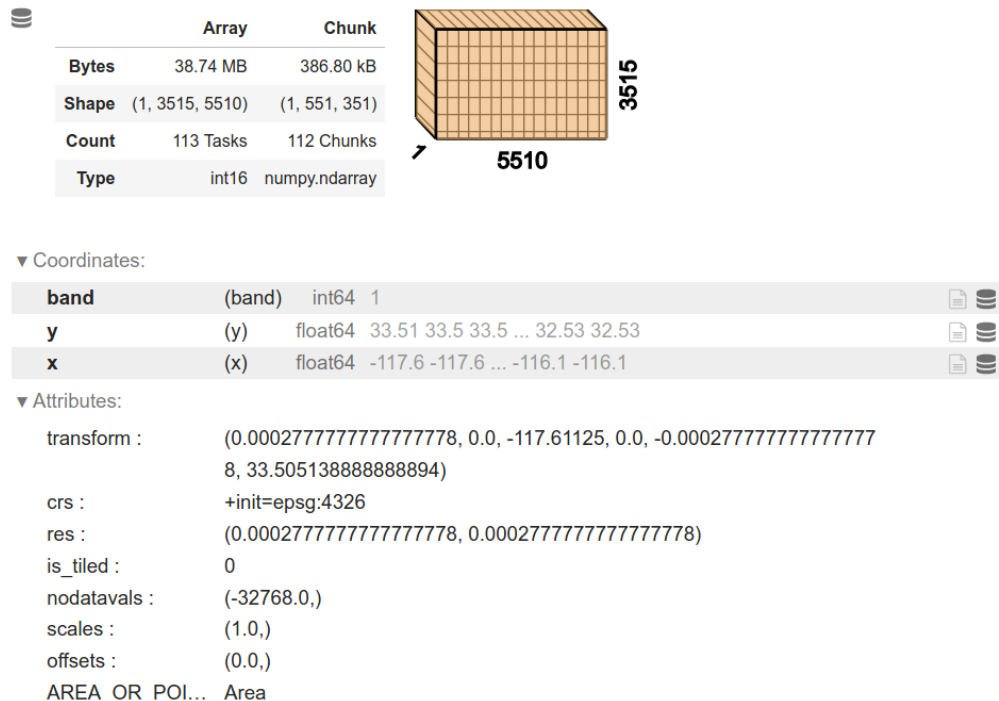
- Models are *one* piece of a broader process of simplification & operationalization that turns reality into representations suitable for computers and statistics.
- Most of data and geographical science is empirical, in that it consists of the analysis and interpretation of measurements.
- The *object*, *network*, and *field* discussed in the previous section are usually still too abstract to help us in this context.
 - So, we pair these data models with other constructs that tell us how quantities can be stored in computers and made amenable for analysis.
 - This is the role that “data structures” play.



Data Structures

- Data structures are digital representations that connect data models to computer implementations.
 - e.g. “geotables”, GeoDataFrames, sf data frames, etc.

xarray.DataArray (band: 1, y: 3515, x: 5510)

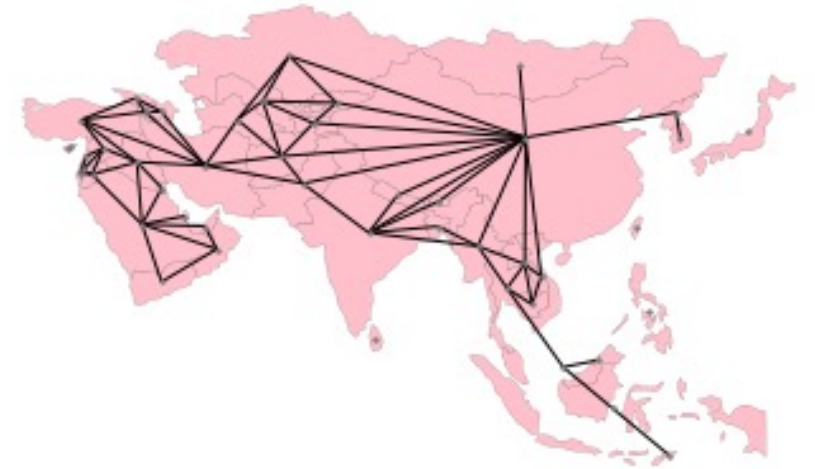


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1	53950935	Africa	Tanzania	TZA	150600.0	POLYGON ((33.90371 -0.95000, 34.07262 -1.05982...
2	603253	Africa	W. Sahara	ESH	906.5	POLYGON ((-8.66559 27.65643, -8.66512 27.58948...
3	35623680	North America	Canada	CAN	1674000.0	MULTIPOLYGON (((-122.84000 49.00000, -122.9742...
4	326625791	North America	United States of America	USA	18560000.0	MULTIPOLYGON (((-122.84000 49.00000, -120.0000...
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Spatial graphs

- Spatial graphs capture relationships between objects that are mediated through space. In a sense, they can be considered geographic networks, a data structure to store topologies.
- Spatial graphs record information about how a given observation is spatially connected to others in the dataset.
- The term spatial graph is sometimes interchangeably used with that of *spatial network*.



Connecting the conceptual to the computational

- Now that the distinction between the conceptual data model and computational data structure is clear, we should explore the ways in which these are traditionally aligned.
- The main conceptual mapping of data model to data structure is inherited from advances made in computer graphics.
 - This traditional view represents fields as rasters and objects as vector-based tables.
 - This categorization is now breaking up!
- Statistical learning methods are getting very good at efficiently translating data between different representations.

